

Megha Pulakhandam

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Software Engineer - Distributed Systems & Cloud | Spring Boot, Node.js | Docker, AWS, Azure

Education

University of Colorado Boulder

Boulder, CO

Master's in Computer Science | 3.8/4.0

Aug 2024 - May 2026

- **Coursework:** Datacenter Scale Computing, Database systems, NLP, Big Data Architecture, Software Engineering

SRM University

Chennai, India

Bachelor's in Computer Science and Engineering | 3.9/4.0

Aug 2020 - May 2024

- **Coursework:** Data structures and algorithms, Database systems, Operating systems, Computer Networks, Cloud Computing, Automata Theory, Software Engineering, Machine Learning, Deep Learning, NLP, Computer Vision

Experience

BioCollate - Software Engineer - Distributed Systems - Colorado, USA

Aug 2025 - Present

- Architected a secure real-time **P2P messaging system** using Signal-style cryptography (X25519, Ed25519, Double Ratchet), enabling 100% end-to-end encrypted communication while maintaining forward secrecy and post-compromise recovery.
- Engineered a **3-tier distributed architecture (Flutter client, Node.js gateway, Tendermint DLT)** delivering less than 500ms median message latency with WebSocket transport, offline messaging through 50+ rotating prekey bundles per user.
- Implemented -side key lifecycle and session state management using secure local storage (Isar), handling per-message ratcheting keys and identity credentials while ensuring 0 plaintext exposure to backend services across the entire system.

Infosys (SRM) - Software Engineer Intern - Prototype App Development - Chennai, India

Apr 2023 - Jun 2023

- Engineered a prototype parking management system with workflows and backend allocation logic, achieving 40% faster slot assignment and 25% higher utilization in controlled test scenarios
- Architected **IoT-based event tracking and automated billing pipelines** for the prototype, improving fault detection and demonstrating scalable enhancements to system reliability and user experience

AnsrCoach - Software Development Engineer Intern - Remote, India

Jan 2023 - Feb 2023

- Developed and optimized a **cross-platform** education app with **Flutter frontend, Node.js backend, and MongoDB**, enabling real-time data synchronization and reducing API latency by 25%
- Transformed **stakeholder-driven requirements** into scalable features, resulting in a 30% improvement in student engagement and 30% higher user satisfaction, while ensuring maintainable, production-ready codebase design

AjnaAI (SRM) - Software Engineer Intern - Deep Learning & Distributed Systems - Remote, India

Mar 2022 - Apr 2022

- **Led a team of 7** to build a distributed **fire detection system on Jetson Nano hardware**, integrating deep learning with embedded system design to improve detection speed by 40% and system reliability by 25%
- Executed performance optimizations using **AWS Lambda** and established Cypress-based testing pipelines, ensuring fault-tolerant operation and high availability in safety-critical environments

Technical Skills

Programming: TypeScript, Java, JavaScript, C/C++, Go, Python, Linux/Unix, Object-Oriented Programming, Data Structures

Backend: Spring Boot, Node.js, Express, REST API, GraphQL, Microservices, Maven/Gradle, System Design

Frontend: React, Next.js, React Native, Flutter, Redux, Sass, CSS, TailwindCSS, Bootstrap, UI Development

Databases: PostgreSQL, MongoDB, MySQL, DynamoDB, CassandraDB, AWS Aurora, Oracle, Firebase

Cloud/DevOps: Azure, AWS (EKS, IAM, EC2, ECS, S3, RDS, Lambda, Cloudwatch, API Gateway, Redshift), GCP, Docker, Redis, Kubernetes, Kafka, CI/CD (GitHub Actions, GitLab), Snowflake, Terraform, Jenkins

Projects

CodeSketch | Node.js, Socket.IO, Redis, Docker, Cloud Infrastructure, AWS, Jest, CI/CD, React

- Architected enterprise-grade real-time collaboration platform with Redis-backed distributed architecture and horizontal auto-scaling, supporting more than 1000 concurrent WebSocket connections with less than 100ms latency and uptime in production
- Established end-to-end DevOps excellence through containerized Docker deployments, automated CI/CD pipelines via GitHub Actions, comprehensive observability using Prometheus & Grafana, with 85% test coverage, enabling zero-downtime releases

Smart API Rate Limiter with Adaptive Scaling | AWS, Java, Spring Boot, Redis, Docker, Kubernetes, Maven

- Designed a distributed API rate limiting service using the Token Bucket algorithm with Redis-backed shared state (AWS ElastiCache) and Lua-based atomic operations, enabling consistency across multiple instances and supporting 50K requests with less than 5ms.
- Deployed a cloud-native, horizontally scalable architecture on AWS using Docker and Kubernetes (EKS, HPA, ConfigMaps, service orchestration), achieving 99.9% request reliability and protecting backend services from abuse through HTTP-compliant 429 handling and centralized monitoring

Mind-Bloom | TypeScript, Docker, React, Express, PostgreSQL, JWT, Prisma, GCP, TailwindCSS

- Built and deployed a full-stack mental wellness application featuring mood tracking, journaling, and personalized affirmations, resulting in a 45% increase in user engagement and daily journaling consistency
- Programmed a secure authentication system with JWT and role-based access control, merged with PostgreSQL database using Prisma ORM, and deployed on Google Cloud Platform for scalability and 99.9% uptime reliability